

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 8: Why Cell Division Matters: From Baby to Teenager, and Bananas from a Test Tube

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
B	B	B	B	B	D	B	D	D

USE AFTER: PREDICT PHASE — Predict: think-pair-share on how a 3 kg baby becomes a 50 kg teenager

Q1. A baby has a body mass of about 3 kg. The same person is 50 kg as a teenager. Which statement best explains what happened to the cells in the body?

- A. The cells only grew bigger, and their number stayed the same.
- B. The number of cells increased because existing cells divided to make new cells.**
- C. The food eaten turned directly into new cells without any cell dividing.
- D. The cells joined together to form larger cells.

Correct answer: B. Growth needs more cells, and new cells come from existing cells dividing. Cells also enlarge a little, but enlargement alone cannot explain the increase from 3 kg to 50 kg. Food only supplies raw materials and energy.

USE AFTER: OBSERVE PHASE — Jigsaw, Expert group A (Growth): doubling estimate 2, 4, 8, 16...

Q2. A human life starts as one fertilised egg cell. Suppose every cell divides into two cells each time, so the numbers go 1, 2, 4, 8, 16 ... How many cells are there after 10 rounds of division?

- A. 512
- B. 1024**
- C. 100
- D. 20

Correct answer: B. The number doubles each round, so after 10 rounds there are $2 \times 2 \times \dots$ (10 times) = $2^{10} = 1024$ cells. 20 and 100 come from adding or squaring instead of doubling, and 512 is the number after only 9 rounds.

USE AFTER: OBSERVE PHASE — Jigsaw, Expert group B (Repair and replacement): footballer in Kakamega healing a cut

Q3. A footballer in Kakamega falls and gets a cut on his knee. After a few days the cut has healed. Which statement best explains how the skin repaired itself?

- A. Skin cells next to the cut divided by meiosis to make new skin cells.
- B. Skin cells next to the cut divided by mitosis to make new, identical skin cells.**
- C. The food he ate changed directly into new skin.
- D. The skin cells next to the cut only became larger to fill the gap.

Correct answer: B. Repair and replacement of body cells happens by mitosis, which makes genetically identical cells. Meiosis makes gametes, and cells cannot repair a cut by enlarging alone.

USE AFTER: OBSERVE PHASE — Jigsaw, Expert group C (Reproduction and development): meiosis making gametes

Q4. Which type of cell division produces gametes (sex cells) for sexual reproduction and also leads to variation among offspring?

- A. Mitosis
- B. Meiosis**
- C. Binary fission
- D. Tissue culture

Correct answer: B. Meiosis produces gametes and gives genetic variation. Mitosis makes identical cells for growth and repair, and binary fission is how bacteria reproduce.

USE AFTER: OBSERVE PHASE — Jigsaw, Expert group D (Tissue culture): why a tissue-cultured banana plantlet matches its parent

Q5. A tissue-cultured banana plantlet looks exactly like the parent banana plant it came from. What is the best reason?

- A. The plantlet was grown in the same soil as the parent plant.
- B. Its cells were made by repeated mitosis, which produces genetically identical cells.**
- C. The nutrient medium in the jar changes the plantlet to look like the parent.
- D. Its cells were made by meiosis, which mixes the genes of both parents.

Correct answer: B. Tissue culture multiplies plant cells by repeated mitosis. Mitosis copies the genetic information, so the plantlets are clones of the parent. Meiosis would cause variation.

USE AFTER: EXPLAIN PHASE — Discussion diamond: 'Tissue culture plants are all identical, so one disease could affect them all'

Q6. Tissue-cultured banana plants are all identical to the parent plant. A farmer in Meru worries about this. What is the most sensible concern?

- A. The plants will each divide by meiosis and become different from each other.
- B. The plants will not be able to grow roots.
- C. The plants cannot get their food from the nutrient medium.
- D. Because the plants have the same genes, one disease could affect all of them.**

Correct answer: D. Clones have the same genetic make-up, so they share the same weaknesses. A disease that attacks one plant can attack all of them. This is a real limitation of cloning.

USE AFTER: DQB PHASE — Driving Question Board update: 'How does the body produce the enormous number of new cells needed for growth?' (food versus cell division caption)

Q7. Complete the sentence: 'Food supplies ____, and cell division supplies ____.'

- A. energy only; chromosomes
- B. raw materials and energy; more cells**
- C. raw materials; bigger cells only
- D. new cells; raw materials and energy

Correct answer: B. Food (like ugali and sukuma wiki) provides the raw materials, such as amino acids, glucose and minerals, and the energy to build cells. Cell division is what increases the number of cells, because new cells come from existing cells.

USE AFTER: MODEL BUILDING PHASE — Model revision: side panel showing tissue culture (explant to callus to plantlets to hardened banana or tea plants)

Q8. A group draws the tissue culture side panel of its model poster for tea plants from Kericho. Which order of steps is correct?

- A. Hardening off, explant, callus, shoots, roots
- B. Explant, roots, shoots, callus, hardening off
- C. Callus, explant, roots, shoots, hardening off
- D. Explant, callus, shoots, roots, hardening off**

Correct answer: D. A piece of plant (explant) is placed in sterile nutrient medium and divides by mitosis to form a callus. The callus forms shoots, then roots, and finally the plantlets are hardened off before planting out.

USE AFTER: END OF LESSON — Model check and exit ticket: 'What does mitosis do to the chromosomes?' (end of lesson)

Q9. In a cell that divides by mitosis, what happens to the chromosomes so that the two new cells are genetically identical to the parent cell?

- A. The chromosomes are not copied, so one new cell gets all of them.
- B. The chromosomes are halved, so each new cell gets half of the parent's chromosomes.
- C. The chromosomes are shared out randomly, so each new cell gets a different set.
- D. The chromosomes are copied, and each new cell receives a full set identical to the parent's.**

Correct answer: D. Before mitosis the genetic material is copied. Each daughter cell then receives a complete set identical to the parent's. Halving the chromosome number happens in meiosis, not mitosis.

